A Multi-Level Data-Driven Analysis of Road Traffic Accidents in India

Geographical, Temporal, Seasonal, Environmental and Severity Assessment

B.Tech Research Report — 2026 (conventional format; department template unavailable)

Certificate / Declaration / Acknowledgement — placeholders for candidate completion.

Abstract

Background: India recorded 487,707 accidents and 177,175 deaths in 2024 (+3.7%/+12.4% since 2018) with severity rising 33.50 -> 36.33 deaths/100 accidents, yet exposure denominators are missing. Objective (RQ1-RQ11): relate geography, time, weather, road-user, violation, and collision factors to occurrence and severity using verified sources. Methods: MoRTH 2018-2024 panel (254 state-years, 38 entities); conditional AR2017 2014-2017 candidate; NCRB ADSI 2024 monthly (separate vintage); IMD rain/temp to states (236 matched of 254 state-years across 38 reporting entities); AR-2024 supporting tables; OLS/corr/chi2/Friedman + BH-FDR; lag-only ML (train 2019-22 N=141; test 2023-2024 N=72) vs naive baseline (acc RMSE 1038.6); K-means (K=2-6). Results: pooled rain r=-0.234/-0.376, Tmax r=+0.452/+0.498, Tmin r=+0.366/+0.340 (associational); severity-weather null; Friedman p=3.0e-4 (descriptive zone differences); NCRB May 8.84%/Jul 7.46%; no trained model beat persistence (acc RMSE 1038.6 vs 2148.6 best trained); K=2 (silhouette 0.356). Conclusion: severity rises while counts fluctuate; risk ranking, causal, and forecasting claims are unsupported; state vehicle stock and microdata are binding gaps. Keywords as above.

Abbreviations: MoRTH, NCRB, ADSI, IMD, CMPG, ABDB, RGI, BRSI, TRW, OLS, FDR, BH, RMSE, MAE, KNN (supervised).

Chapter 1 — Introduction

India 487,707 accidents / 177,175 deaths (2024); severity 36.33; RQ1-RQ11 observational; objectives 1-13; scope 2018-2024 benchmark + conditional layers.

Chapter 2 — Literature Review

LITERATURE REVIEW SOURCES REQUIRED — no external literature collected; nothing invented.

Chapter 3 — Research Methodology

Per PHASE10_METHODOLOGY.md: verification by reconciliation; OLS/corr/chi2/Friedman + BH-FDR; lag-only ML vs naive; K-means K=2; seed 42.

Chapter 4 — Data Sources and Validation

254 state-years, 38 entities, 0 mismatches; NCRB separate (-19,740); IMD 236/18; 7 modules VERIFIED-MIRROR; TN reconciled; AR2017 conditional 12/12.

Chapter 5 — India-Level Analysis

+3.7% acc / +12.4% fat; 2020 dip; slopes ns (N=7).

Chapter 6 — Geographical Analysis

Friedman chi2=25.29 p=3.0e-4 W=0.72 (descriptive); state slopes mostly non-detectable; no rankings.

Chapter 7 — Temporal and Seasonal Analysis

Annual fluctuation; NCRB May 8.84% / July 7.46% (NCRB vintage, never merged).

Chapter 8 — Environmental Analysis

Rain -0.234/-0.376; Tmax +0.452/+0.498; Tmin +0.366/+0.340 (FDR, N=236); severity null; never causal.

Chapter 9 — Severity, Cause, Collision, Road-User

Over-speeding 70.3% (reported); Hit from Back leads; two-wheelers 46.2% (counts); licence/safety reported shares.

Chapter 10 — Statistical Analysis

Full methods + FDR + effect sizes; near-zero rule documented.

Chapter 11 — Machine Learning and Clustering

Naive RMSE 1038.6/286.3 beats all trained (honest null); K=2 silhouette 0.356 descriptive.

Chapter 12 — Discussion

Per PHASE10_DISCUSSION_SECTION.md.

Chapter 13 — Limitations

Vehicle stock, microdata, monthly MoRTH, N=213, vintages (full matrix).

Chapter 14 — Contributions

10 bounded contributions.

Chapter 15 — Application Vision

Mode A/B platform; analytical tool, NOT proven predictor.

Chapter 16 — Conclusion

Per PHASE10_CONCLUSION.md.

References

Per FINAL_REFERENCES.md (mirrors labeled; nothing invented).

Appendices

A dictionary; B availability; C statistics; D ML; E clustering; F graphs; G reproducibility.

Figure F01. F01. Annual road accidents in India, 2018-2024 (verified MoRTH benchmark). Recorded totals

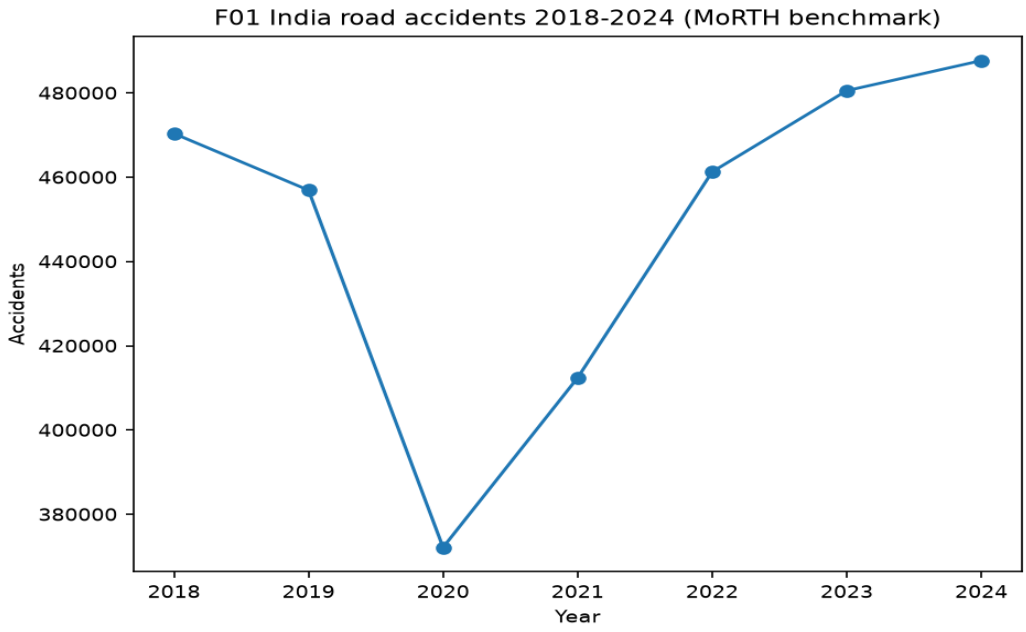


Figure F02. F02. Annual road fatalities in India, 2018-2024 (MoRTH benchmark). Fatalities rose 12.4%, faster

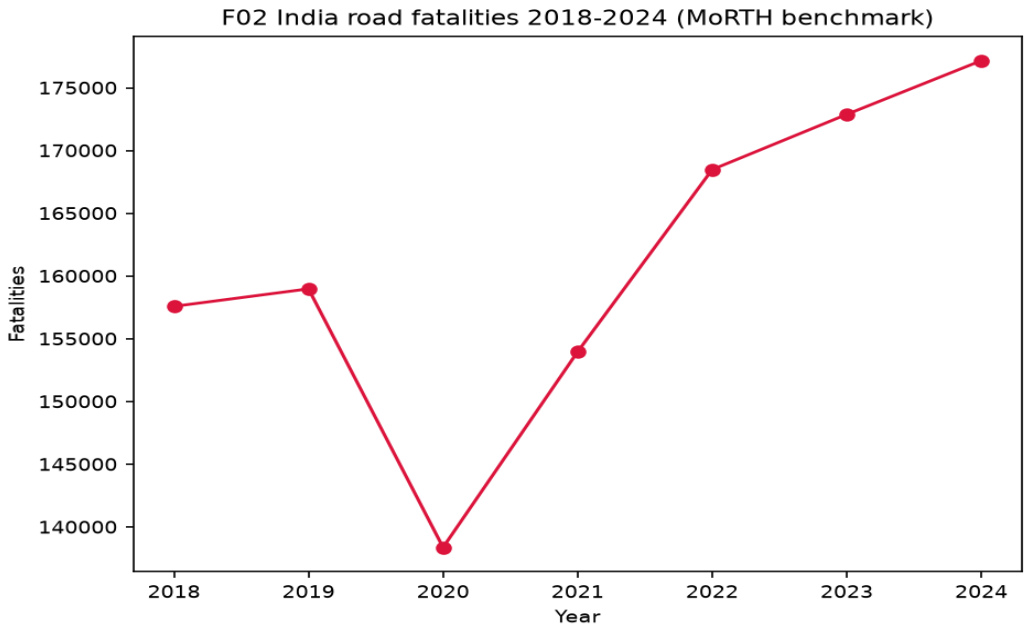


Figure F03. F03. Fatality ratio, deaths per 100 accidents, 2018-2024 (derived). Rose 33.50 to 36.33; a

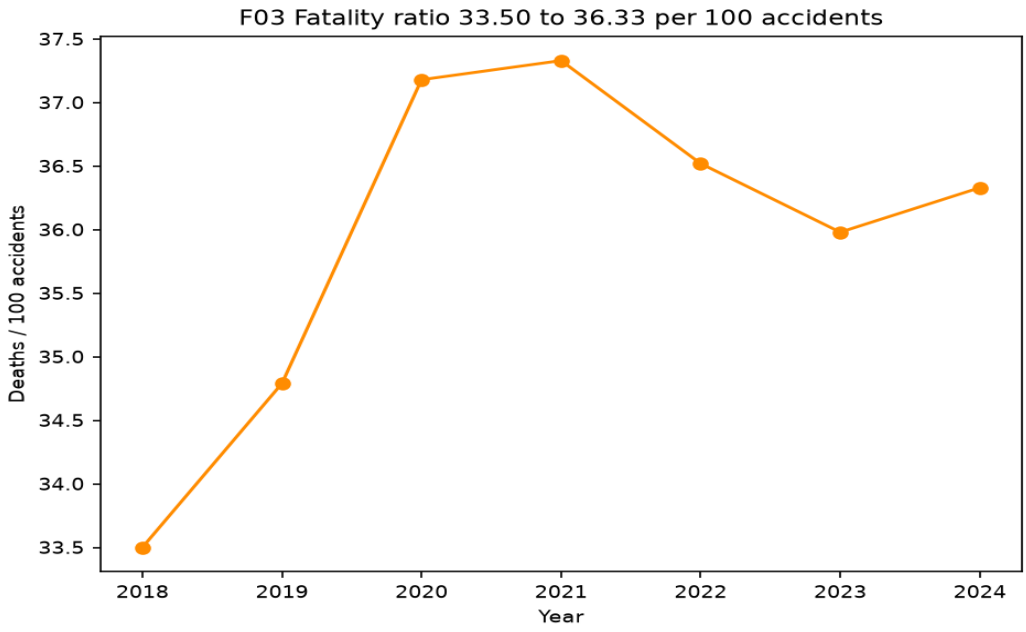


Figure F04. F04. Zone accident trends by six zones (benchmark + mapping). Descriptive differences only; no ranking.

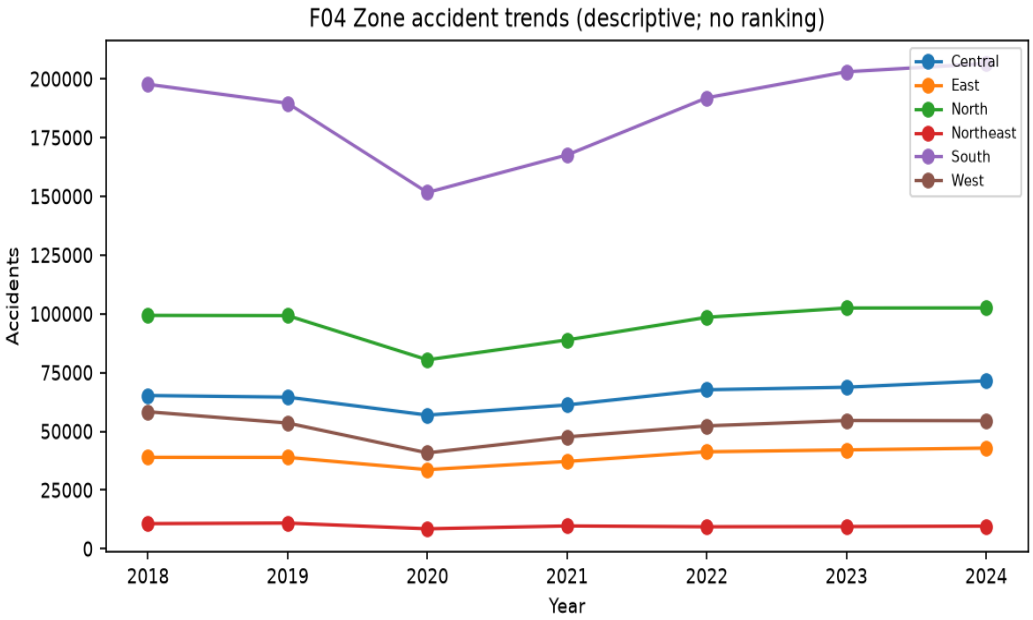


Figure F07. F07. State-year rainfall vs accidents (IMD join, N=236). Pooled r=-0.234; scale confounding likely; not causal.

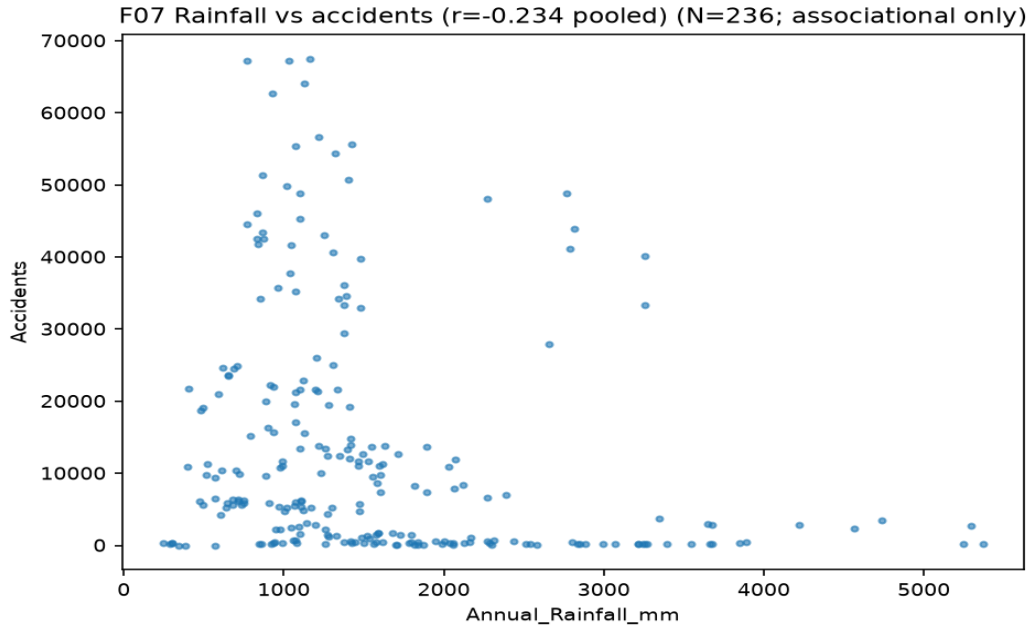


Figure F08. F08. State-year mean Tmax vs fatalities (N=236). Pooled r=+0.498; associational only.

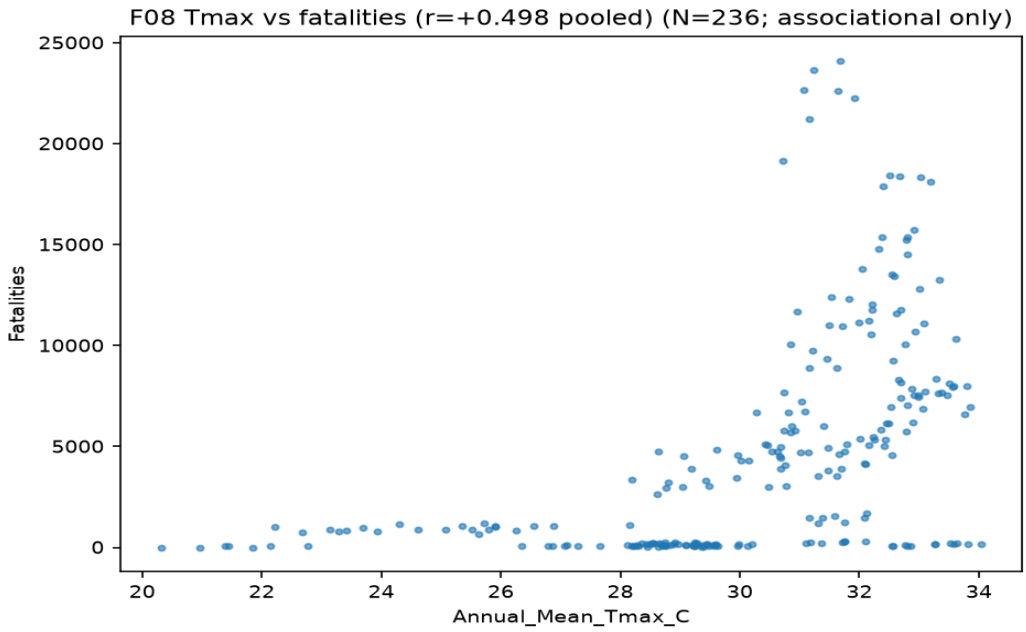


Figure F09. F09. 2024 fatalities by reported violation category (MoRTH AR2024 mirror, India-level). Over-speeding

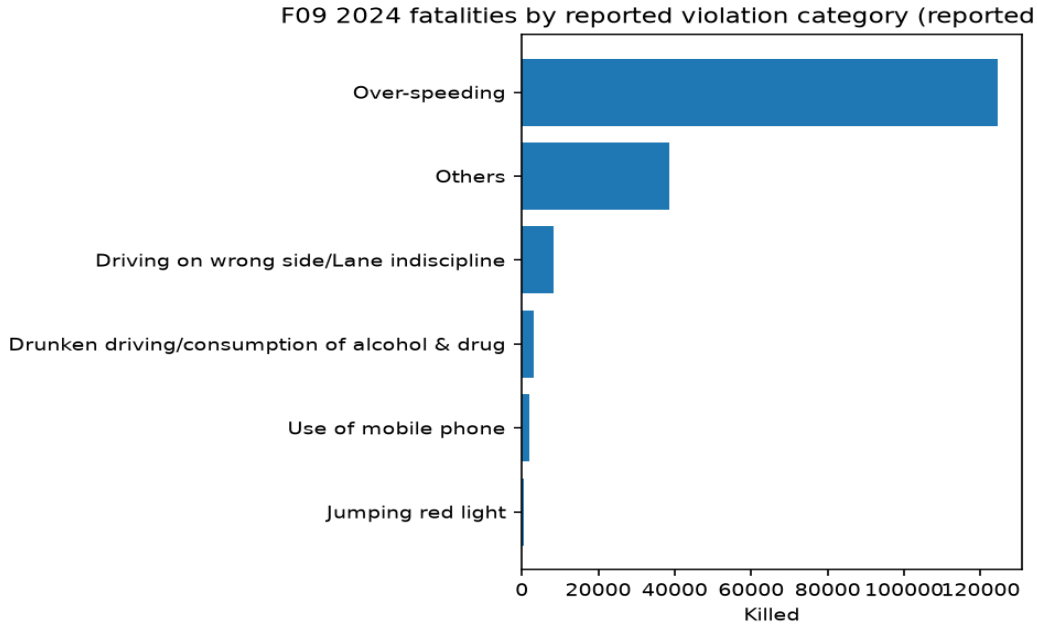


Figure F11. F11. 2024 fatalities by road-user mode (AR2024 mirror). Counts without exposure; largest count is not highest risk.

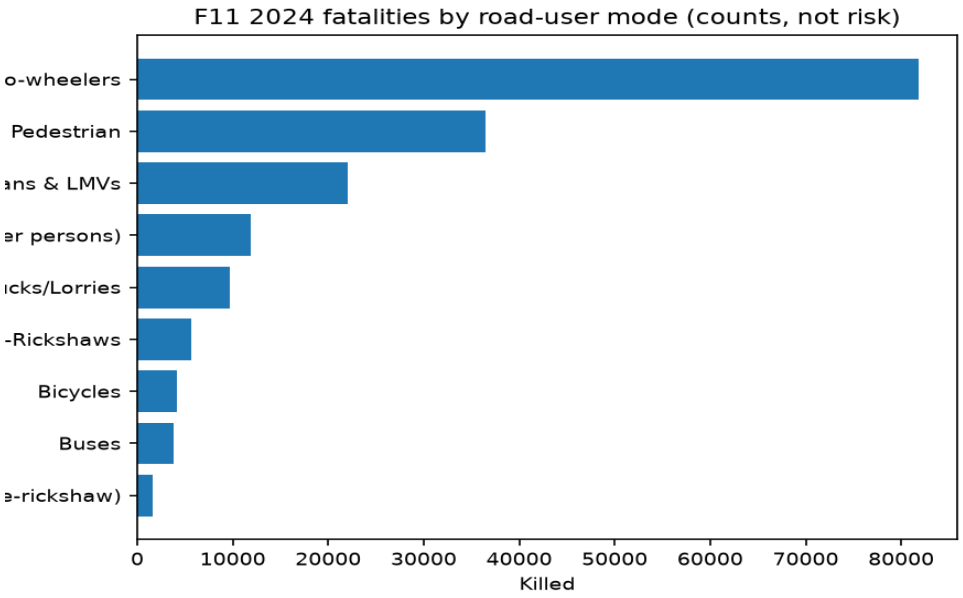


Figure F13. F13. 2024 monthly road accidents, NCRB ADSI Table 1A.5 Road-States (NCRB vintage, NOT MoRTH). May peak, July trough.

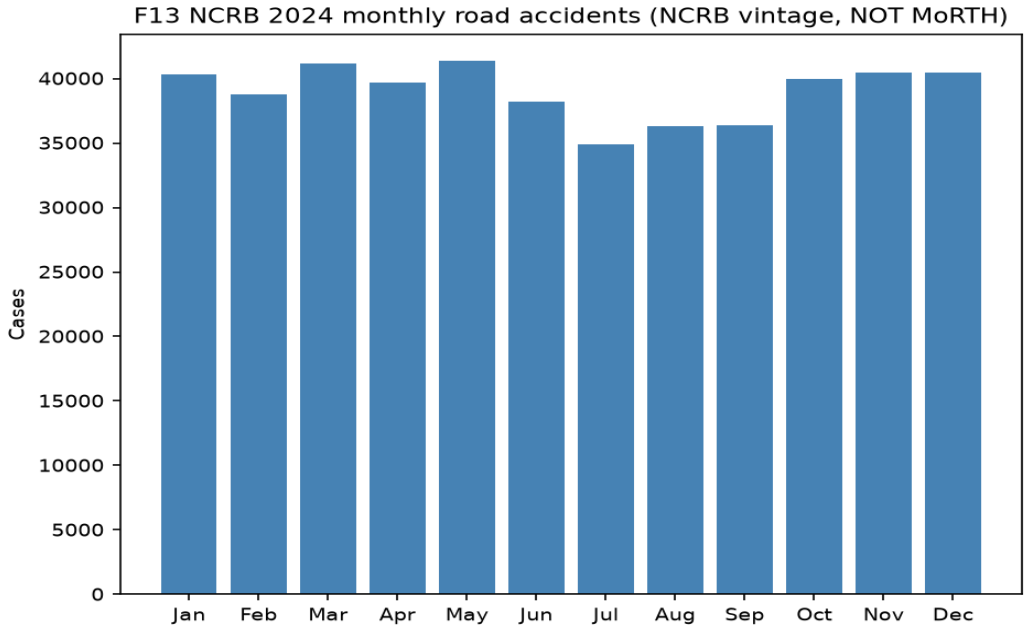


Figure F15. F15. Actual vs predicted accidents with test-RMSE comparison (2023-2024 holdout). Naive persistence

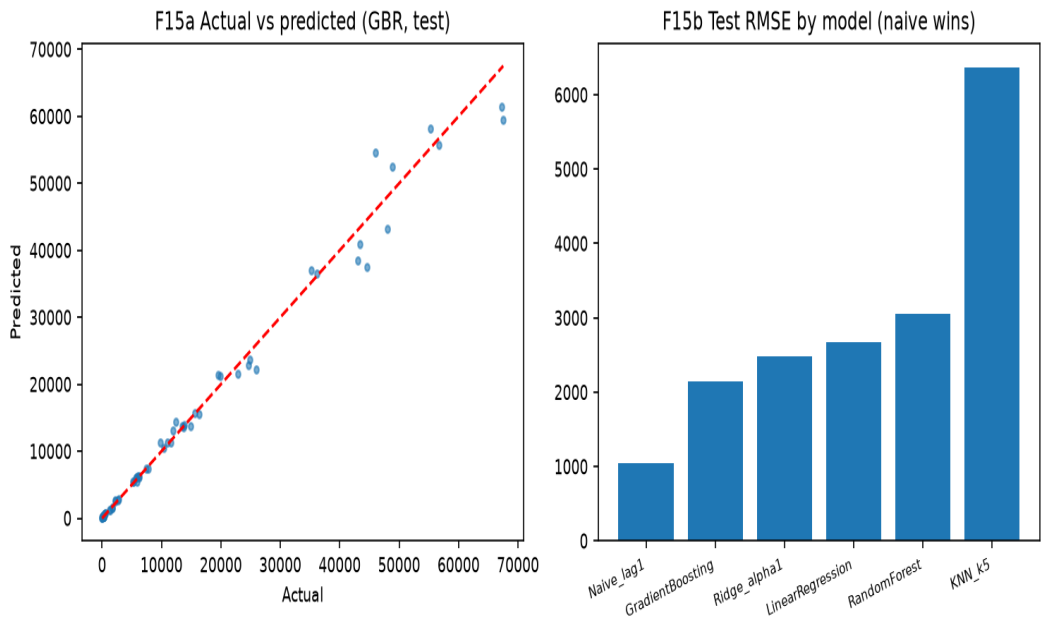
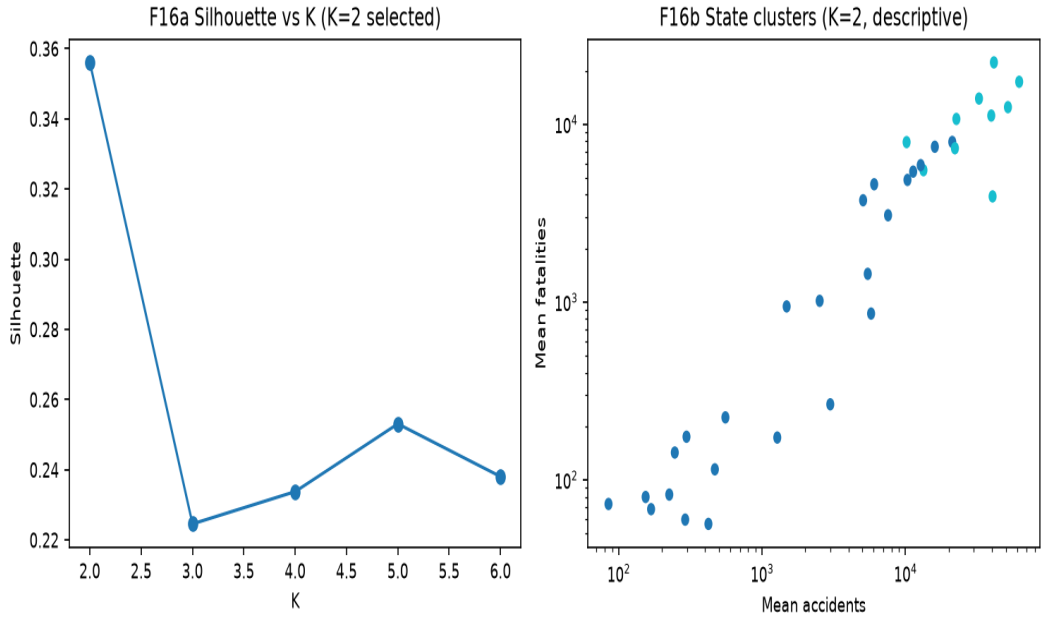


Figure F16. F16. K-means K-selection and PCA cluster scatter (K=2, silhouette 0.356). Exploratory typology; descriptive labels.



Tables 1-15. Per FINAL_TABLES.md (values preserved exactly).